

COSC 650 HW1

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1. Convex Optimization

Consider the problem
$$\min_x e^{x_1} + e^{x_2} + \dots + e^{x_n}$$
$$\text{s.t. } x_1 + x_2 + \dots + x_n = \gamma$$

a. Show that the above optimization problem is convex.

proof.

$\Rightarrow$ Let $f(x) = \sum_{i=1}^n e^{x_i}$ w/ feasible set $C = \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = \gamma\}$

$x \rightarrow x_i$ is affine

e^t is convex on $\mathbb{R}$

} this means the composition $x \rightarrow e^{x_i}$ is convex

because a non-negative sum of convex functions is convex

$f(x)$ is convex on $\mathbb{R}^n$

$\Rightarrow$ the constraint can be written as $1^T x = \gamma$ where $1 = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$

Since $x \rightarrow 1^T x$ is affine, the equality constraint defines an affine set. Every affine set is convex so C is convex.

$\therefore f(x)$ w/ feasible set C is a convex optimization problem

b. Use the equation $[x_n = \gamma - x_1 - \dots - x_{n-1}]$ to rewrite the above optimization problem as an unconstrained optimization problem.

Sol.

$\Rightarrow$ using the equality constraint $x_n = \gamma - \sum_{i=1}^{n-1} x_i$

Sub the expression into the objective func $\sum_{i=1}^n e^{x_i} = \sum_{i=1}^{n-1} e^{x_i} + e^{(\gamma - \sum_{i=1}^{n-1} x_i)}$

$\therefore$ the original constrained optimization is no longer constrained

$$\min_{(x_1, \dots, x_{n-1}) \in \mathbb{R}^{n-1}} \left[\sum_{i=1}^{n-1} e^{x_i} + e^{(\gamma - \sum_{i=1}^{n-1} x_i)} \right]$$

c. Find the set of global optimum points

sol. note. because this obj is differentiable and convex,
any stationary point is a global minimum.

$$\text{Let } f(x_1, \dots, x_{n-1}) = \sum_{i=1}^{n-1} e^{x_i} - e^{\gamma - \sum_{i=1}^{n-1} x_i}$$

$$\text{for each } i=1, \dots, n-1 \quad \frac{df}{dx_i} = e^{x_i} - e^{\gamma - \sum_{j=1}^{n-1} x_j}$$

$$\text{at a stationary point } \frac{df}{dx_i} = 0 \text{ so } e^{x_i} = e^{\gamma - \sum_{j=1}^{n-1} x_j}$$

$$\log(\cdot) \text{ leaves } x_i = \gamma - \sum_{j=1}^{n-1} x_j \text{ the RHS is } x_n, x_i = x_n$$

$$\text{the original constraint now reads } x_1 = x_2 = \dots = x_n = \gamma$$

$$\text{let } C \text{ be the common value and } n \text{ be the total } nC = \gamma \quad C = \frac{\gamma}{n}$$

$$x_1^* = x_2^* = \dots = x_n^* = \frac{\gamma}{n}$$

$$\therefore x^* = \left(\frac{\gamma}{n}, \dots, \frac{\gamma}{n} \right)$$

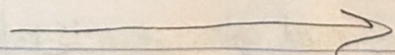
d. Apply the change of variables $y_i = e^{x_i}$ to prove the arithmetic-geometric mean inequality

$$\frac{1}{n} \sum_{i=1}^n y_i \geq \left(\prod_{i=1}^n y_i \right)^{\frac{1}{n}} \quad \forall (y_1, \dots, y_n) \in \mathbb{R}_+^n$$

sol. from 1c we know $x^* = \left(\frac{\gamma}{n}, \dots, \frac{\gamma}{n} \right)$ so for every feasible x ,

$$\sum_{i=1}^n e^{x_i} \geq n e^{\frac{\gamma}{n}} \quad \left(n e^{\frac{\gamma}{n}} \text{ since min it will always be less than or equal to } \sum \right)$$

$$\text{let } y_i = e^{x_i} > 0 \text{ so } x_i = \log y_i$$



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original constraint

$$\gamma = \sum_{i=1}^n \log y_i = \log \left(\prod_{i=1}^n y_i \right)$$

$$\frac{\gamma}{n} = \frac{1}{n} \log \left(\prod_{i=1}^n y_i \right) \rightarrow \frac{\gamma}{n} = \log \left(\left(\prod_{i=1}^n y_i \right)^{\frac{1}{n}} \right)$$

raise to e: $e^{\frac{\gamma}{n}} = \left(\prod_{i=1}^n y_i \right)^{\frac{1}{n}}$

multiply by n: $ne^{\frac{\gamma}{n}} = n \left(\prod_{i=1}^n y_i \right)^{\frac{1}{n}} \rightarrow \sum_{i=1}^n e^{x_i} \geq n \left(\prod_{i=1}^n y_i \right)^{\frac{1}{n}}$

sub y_i for e^{x_i} and divide by n: $\therefore \frac{1}{n} \sum_{i=1}^n y_i \geq \left(\prod_{i=1}^n y_i \right)^{\frac{1}{n}}$

2. Convexity proofs Prove the following statements

a. The function

$$f(x) = \log(e^{a_1^T x + b_1} + \dots + e^{a_n^T x + b_n}) \text{ is convex}$$

sol.

Let $z_i = a_i^T x + b_i$

$$S = \sum_{i=1}^n e^{z_i} \quad \text{and} \quad p_i = \frac{e^{z_i}}{S} > 0 \quad \& \quad \sum_{i=1}^n p_i = 1$$

$$f(x) = \log S$$

diff: $\nabla f(x) = \frac{1}{S} \nabla S$

note: $\nabla e^{a_i^T x + b_i} = e^{a_i^T x + b_i} a_i$

so $\nabla S = \sum_{i=1}^n e^{z_i} a_i \rightarrow \nabla f(x) = \frac{\sum_{i=1}^n e^{z_i} a_i}{\sum_{i=1}^n e^{z_i}}$

gradient becomes $\nabla f(x) = \sum_{i=1}^n p_i a_i$

get hessian:

$$\nabla^2 f(x) = \sum_{i=1}^n p_i a_i a_i^T - \left(\sum_{i=1}^n p_i a_i \right) \left(\sum_{i=1}^n p_i a_i \right)^T$$

to show positive semidefinite. take any vector $v \in \mathbb{R}^m$, then

$$v^T \nabla^2 f(x) v = \sum_{i=1}^n p_i (v^T a_i)^2 - \left(\sum_{i=1}^n p_i (v^T a_i) \right)^2 \text{ this is a weighted variance}$$

$$v^T \nabla^2 f(x) v \geq 0 \quad \therefore \nabla^2 f(x) \succeq 0 \text{ (positive semidefinite)}$$

$f(x)$ is convex

WTS: Each level of composition is convex

b. Let $\sigma_1(Z)$ denote the largest singular value of the matrix Z .

Show that the function $f(x) = \exp((\sigma_1(Ax))^{10})$ is convex.

Sol. note. This is handled using composition rules for convex func

$\Rightarrow \sigma_1(Z) = \|Z\|_2$ spectral norm is convex

$\Rightarrow X \rightarrow AX$ is a linear mapping (Affine) so $g(x) = \sigma_1(Ax)$ is convex

$\Rightarrow$ Singular Values are always non-negative $g(x) = \sigma_1(Ax) \geq 0$

consider $h(t) = t^{10} \rightarrow h'(t) = 10t^9 \rightarrow h''(t) = 90t^8$

for $t \geq 0$ both derivatives are $[0, \infty)$ both convex and

non-decreasing so $h(g(x)) = (\sigma_1(Ax))^{10}$ is convex

$\Rightarrow \exp()$ is convex and strictly increasing so convexity is preserved in $f(x) = \exp((\sigma_1(Ax))^{10})$

$\therefore f(x)$ is convex

3. (Stationary Points) for each value of the scalar β , find the set of all stationary points of the optimization problem

$$\min_{\substack{x, y \\ s.t. (x, y) \in \mathbb{R}^2}} x^2 + y^2 + \beta xy + x + 2y$$

Which of these stationary points are global minima? For an unconstrained smooth optimization problem, a point is stationary if the gradient of the objective at that point is 0.

Sol.

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WTS: stationary points and which are global minima

$$\Rightarrow \frac{df}{dx} = 2x + \beta y + 1 \quad \frac{df}{dy} = 2y + \beta x + 2 \quad \nabla f(x, y) = \begin{bmatrix} 2x + \beta y + 1 \\ \beta x + 2y + 2 \end{bmatrix}$$

$$\begin{cases} 2x + \beta y + 1 = 0 \\ \beta x + 2y + 2 = 0 \end{cases} \quad \begin{cases} 2x + \beta y = -1 \\ \beta x + 2y = -2 \end{cases}$$

$$\begin{bmatrix} 2 & \beta \\ \beta & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} \quad \det \begin{bmatrix} 2 & \beta \\ \beta & 2 \end{bmatrix} = 4 - \beta^2$$

$$\begin{aligned} \text{Solve for } x: \quad 2x + \beta y &= -1 \quad \cdot 2 & 4x + 2\beta y &= -2 \\ \beta x + 2y &= -2 \quad \cdot \beta & -\beta^2 x + 2\beta y &= -2\beta \\ (4 - \beta^2)x &= -2 + 2\beta & x^* &= \frac{-2 + 2\beta}{4 - \beta^2} \end{aligned}$$

$$\begin{aligned} \text{Solve for } y: \quad 2x + \beta y &= -1 \quad \cdot \beta & 2\beta x + \beta^2 y &= -\beta \\ \beta x + 2y &= -2 \quad \cdot 2 & 2\beta x + 4y &= -4 \\ (4 - \beta^2)y &= -4 + \beta & y^* &= \frac{-4 + \beta}{4 - \beta^2} \end{aligned}$$

$$\text{Stationary point } (x^*, y^*) = \left(\frac{2\beta - 2}{4 - \beta^2}, \frac{\beta - 4}{4 - \beta^2} \right) \quad \beta \neq -2, 2$$

$$\nabla f(x, y) = \begin{bmatrix} 2x + \beta y + 1 \\ \beta x + 2y + 2 \end{bmatrix} \quad \nabla^2 f(x, y) = \begin{bmatrix} 2 & \beta \\ \beta & 2 \end{bmatrix}$$

Hessian depends on β not x or y

eigenvalues $2 + \beta$ and $2 - \beta$

to be positive definite, both eigenvalues have to be positive

$$\left. \begin{aligned} 2 + \beta > 0 & \quad \beta > -2 \\ 2 - \beta > 0 & \quad \beta < 2 \end{aligned} \right\} -2 < \beta < 2$$

$\therefore$ The global minima is $\left(\frac{2\beta - 2}{4 - \beta^2}, \frac{\beta - 4}{4 - \beta^2} \right)$ where $-2 < \beta < 2$

4. (Linewise local minima) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable. Suppose that x^* is a local minimum of f along every line that passes through x^* ; that is, the function $g(\alpha) = f(x^* + \alpha d)$ has a local minimum at $\alpha = 0$ for every $d \in \mathbb{R}^n$.

a. Show that $\nabla f(x^*) = 0$

Sol. has local min @ $g'(0) = 0$

$$g(\alpha) = f(x^* + \alpha d) \rightarrow g'(\alpha) = \nabla f(x^* + \alpha d)^T d$$

$$\alpha = 0 \rightarrow g'(0) = \nabla f(x^*)^T d = 0 \text{ for every } d$$

Only zero vector can have dot product with every possible vector

$$\therefore \nabla f(x^*) = 0$$

b. Show by example that x^* need not be a local min of f

Hint: consider $f(y, z) = (z - py^2)(z - qy^2)$ $0 < p < q$

Show that $(0, 0)$ is a local minimum of f along every line that passes through $(0, 0)$. Furthermore, if $p < m < q$, then $f(y, my^2) < 0$ when $y \neq 0$, while $f(0, 0) = 0$

Sol.

$$(y, z) = (0, 0) \text{ we get } f(0, 0) = (0 - p(0)^2)(0 - q(0)^2) = 0$$

need to show there exists another 0 on any line through $(0, 0)$

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Consider any line through $(0,0)$ parameterized by
 $(y,z) = (a\alpha, b\alpha)$

$$\text{then } f(a\alpha, b\alpha) = \alpha^2(b - pa^2)(b - ga^2)$$

for a small $|\alpha|$, this is non-negative. if $b=0$, it becomes

$$pga^4\alpha^4 \geq 0 \quad (0,0) \text{ is a local min along all lines}$$

however choose m such that $p < m < g$ with path $z = my^2$

$$f(y, my^2) = \underbrace{(m-p)}_{(-)} \underbrace{(m-g)}_{(+)} \underbrace{y^4}_{(+)} < 0$$

so there are points close to $(0,0)$ with smaller values

$\therefore (0,0)$ is not a local minimum of f

5. Symmetric Rank-one matrix approximation. Let $A \neq 0$ be a symmetric rank-one square matrix;

$$A = x_0 x_0^T \neq 0, \quad x_0 \in \mathbb{R}^n$$

consider,

$$\min_x \frac{1}{2} \|A - xx^T\|_F^2$$

$$\text{s.t. } x \in \mathbb{R}^n$$

a. for any matrix $\|M\|_F^2 = \text{tr}(M^T M)$

AI note: ChatGPT was used to assist in my understanding of this assignment. I took a great deal of time understanding and creating clear written solutions. I stopped at 5 in the spirit of academic honesty even though I had access to more. I will only ever submit something I can repeat.